

- Multi-Asset Investing
- Factor Exposure Investing

The book is categorized in three parts. Part I focuses on the current electronic market environment where we discuss trading algorithms, market microstructure research, and transaction cost analysis. Part II focuses on the necessary mathematical models that are used to construct, calibrate, and test market impact models, as well as to develop single stock and portfolio trading algorithms. The section further discusses volatility and factor models, as well as advanced algorithmic forecasting techniques. Part III focuses on portfolio management techniques and how TCA and market impact can be incorporated into the investment decisions, stock selection, and portfolio construction to improve portfolio performance. We introduce readers to an advanced portfolio optimization process that incorporates market impact and transaction costs directly into the portfolio optimization. We provide insight into how MI factor scores can be used to improve stock selection, as well as a technique that can be used by portfolio managers to decipher broker-dealer black box models. This section concludes with an overview of high frequency trading, and the necessary mathematical knowledge required to develop black box trading models.